

# Artificial Intelligence: Skills, Applications, Careers, and Research - Day 2

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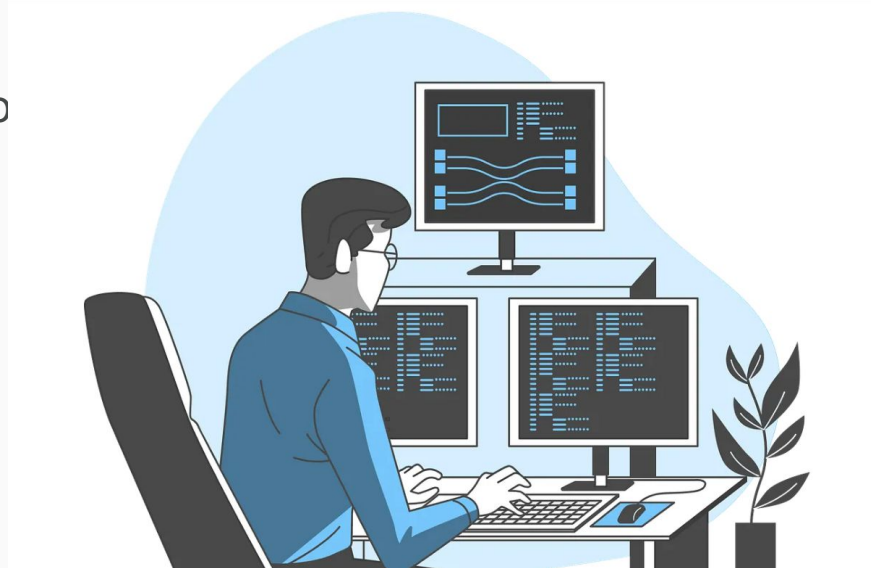
**Leafclutch**  
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# ABOUT SPEAKER

- Visiting Researcher — Deakin University
- Research Affiliate — Monash University
- Peer Reviewer — Journals in Medicine & Health Sciences
- Research focus: Quantum machine learning, Computational neuroscience
- Experience in AI, research, and applied innovation

# Seminar overview

- Why AI is booming
- Why everyone should learn AI now
- Automating tasks using simple Python
- Major branches of AI
- Fields where AI is applied
- AI job opportunities
- Responsibilities of AI roles
- Research conduction process



# Why is AI booming ?

- Explosion of data from digital platforms
- Increased computing power (GPUs, cloud computing)
- Breakthroughs in machine learning and deep learning
- Availability of open-source AI tools
- Industry demand for automation and efficiency
- Investment from governments and private sectors
- Integration of AI into everyday tools (ChatGPT, Copilot, etc.)



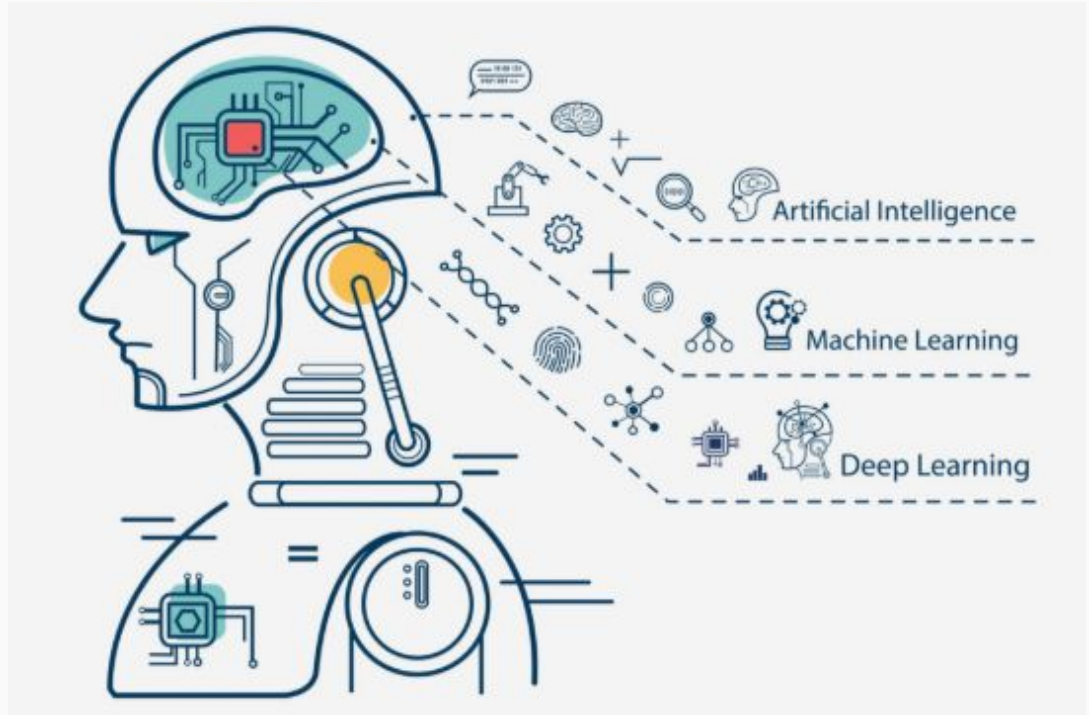
## THE AI BOOM

How Executives Are Embracing AI Tools For Communication



# Why Everyone Needs to Learn AI Now More Than Ever

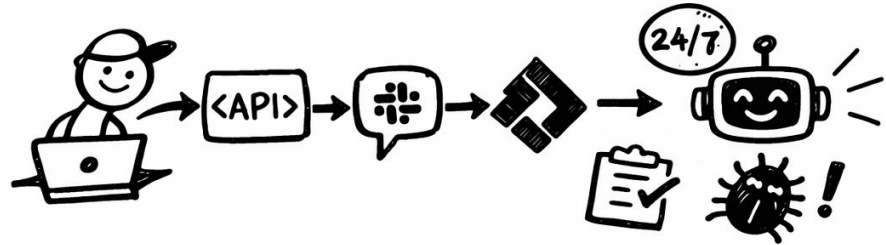
- AI is becoming a core workplace skill
- Automation is reshaping every industry
- AI improves productivity and decision-making
- Competitive advantage in job market
- Enables innovation and entrepreneurship
- AI literacy helps responsible technology use



# Automating Tasks with Just a Few Lines of Python

- Automate repetitive Excel work
- Data cleaning and preprocessing
- Email automation
- File organization
- Report generation
- Web scraping for research data

Automate the Boring Stuff  
You Keep Doing Manually



# Simple Python Automation

- Read data from Excel
- Process and analyze data
- Generate summary
- Export results automatically

## Outcome

- Reduced human error
- Faster workflow
- Scalable solutions

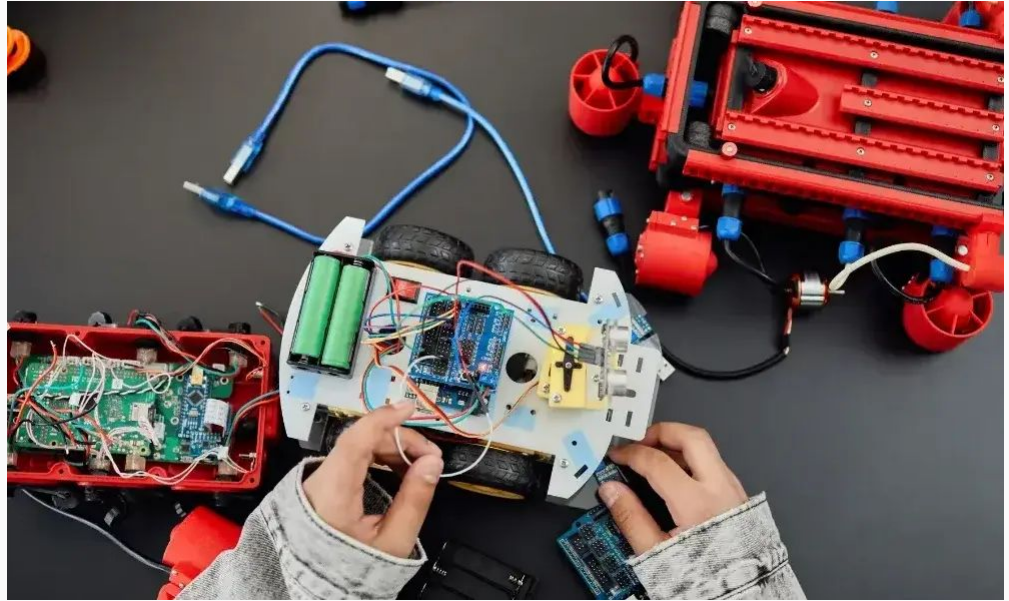


# Major Branches of AI

- Machine Learning
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision
- Robotics
- Expert Systems
- Reinforcement Learning
- Generative AI

# Robotics

- Healthcare & Surgery
- Autonomous Vehicles
- Manufacturing & Industry
- Retail & Customer Service
- Agriculture
- Defence & Security
- Space Exploration



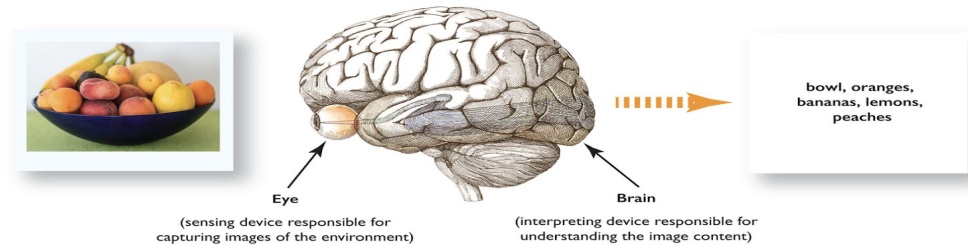
# Computer Vision

- Healthcare
- Security & Surveillance
- Autonomous Vehicles
- Retail
- Manufacturing
- Agriculture

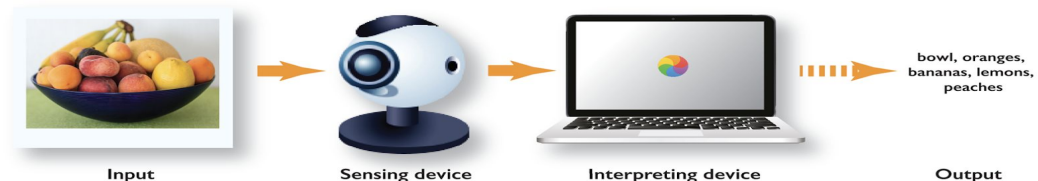
## Human Vision VS Computer Vision



### Human Vision System

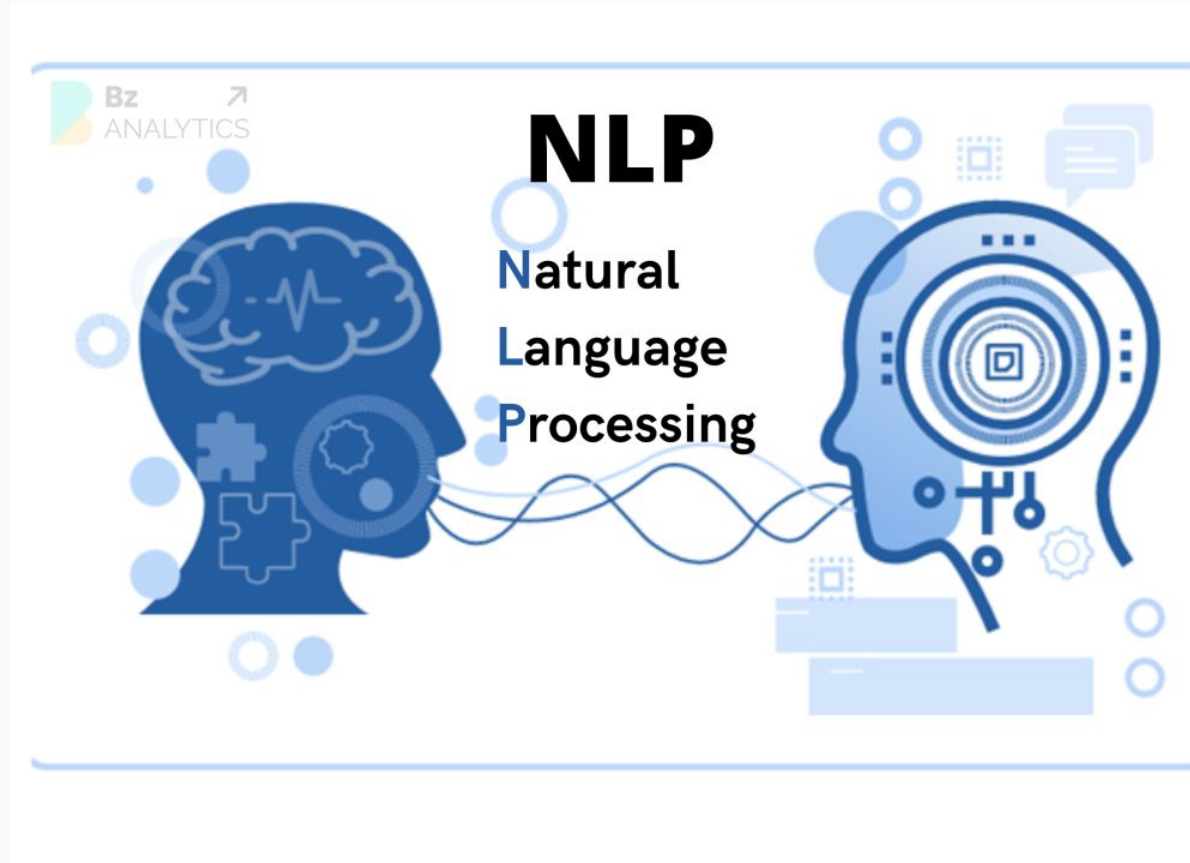


### Computer Vision System



# Natural Language Processing

- Communication & Translation
- Chatbots & Virtual Assistants
- Text Analysis
- Search & Information Retrieval
- Healthcare
- Content Moderation



# Gen Ai

- Content Creation
- Education
- Healthcare
- Business & Marketing
- Design & Creativity
- Gaming & Entertainment
- Software Development

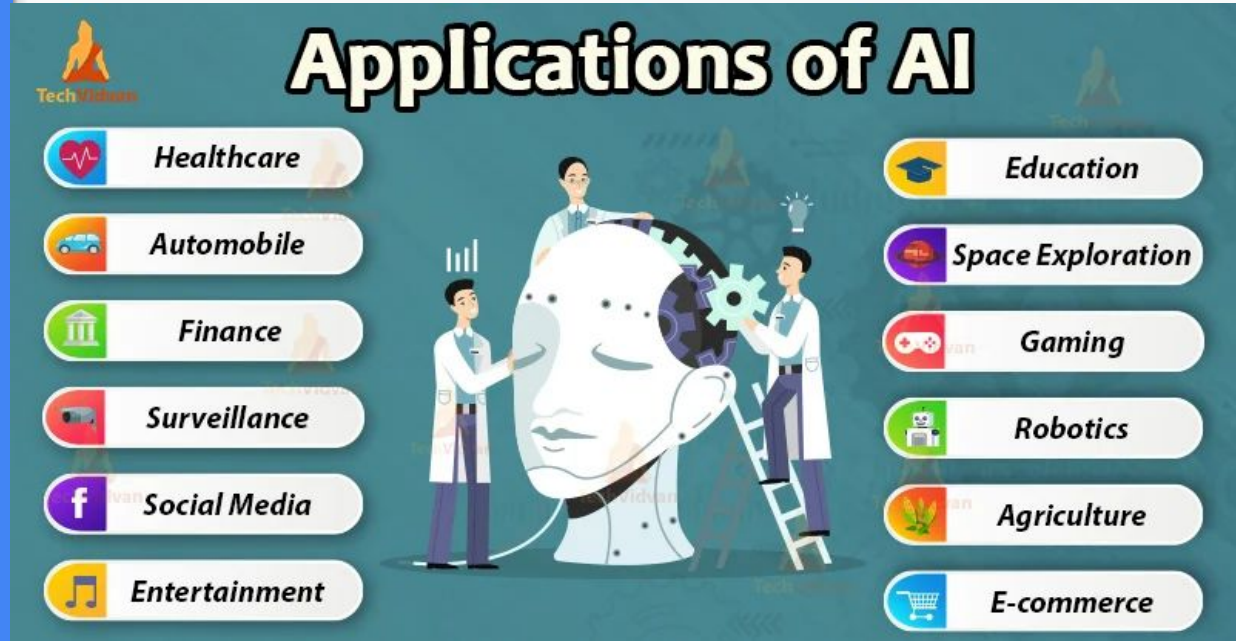
## Generative AI Workflow for Content Creation





# Fields Where AI is Applied

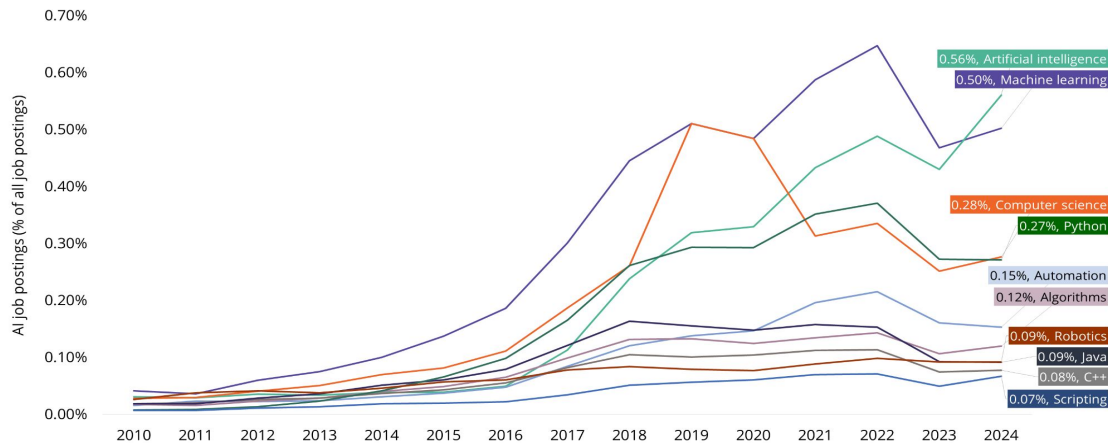
- Healthcare
- Finance
- Education
- Agriculture
- Manufacturing
- Transportation
- Marketing
- Cybersecurity
- Smart Cities
- Research & Academia



# AI Job Opportunities

- AI Engineer
- Machine Learning Engineer
- Data Scientist
- Data Analyst
- AI Researcher
- NLP Engineer
- Computer Vision Engineer
- Robotics Engineer
- AI Product Manager
- AI Consultant

AI job postings (percent of all job postings) in the U.S. by skill cluster



Sources: AVANT by Avison Young, Lightcast

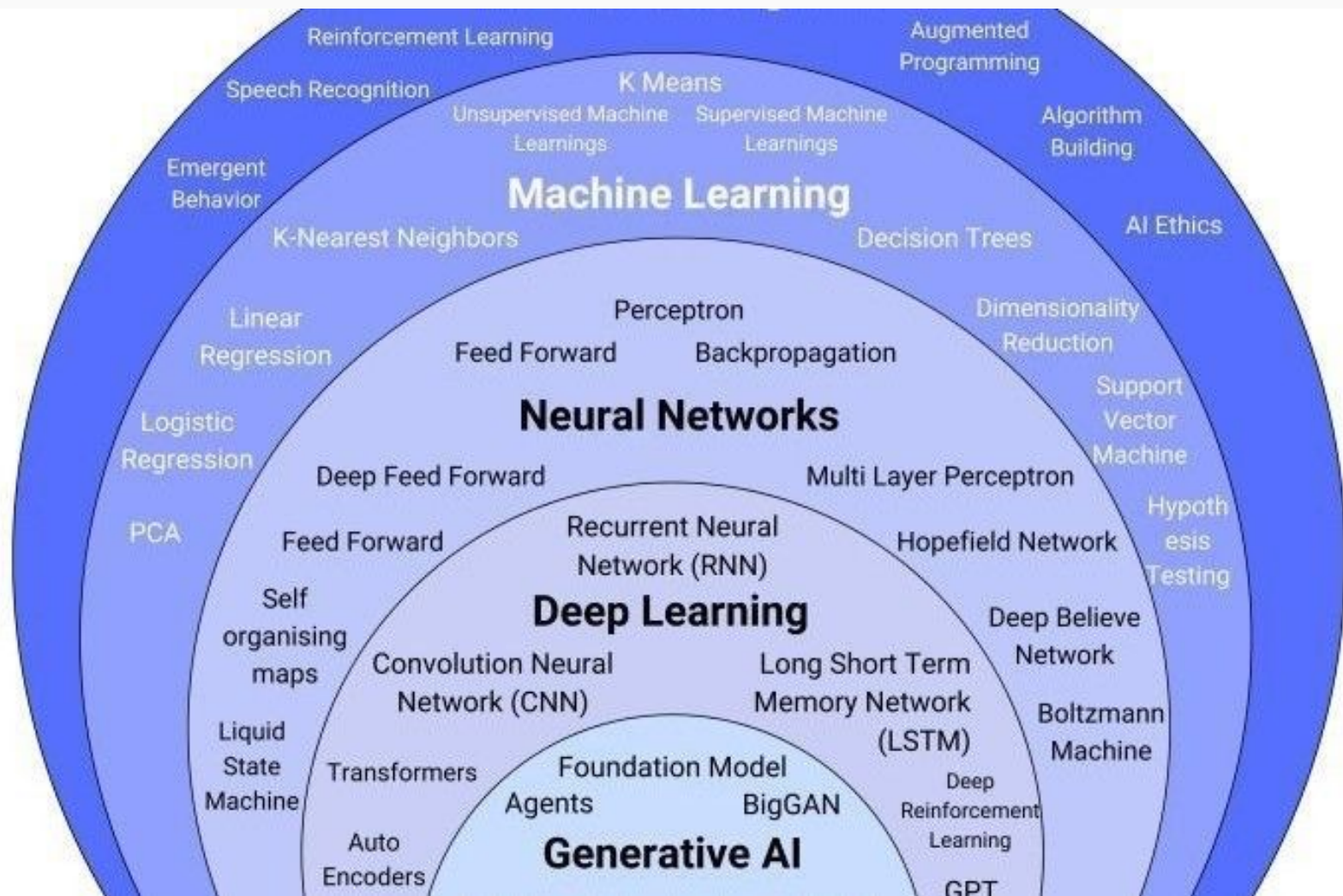
AVANT  
by AVISON YOUNG

CIO

## The 10 most in-demand tech jobs for 2026

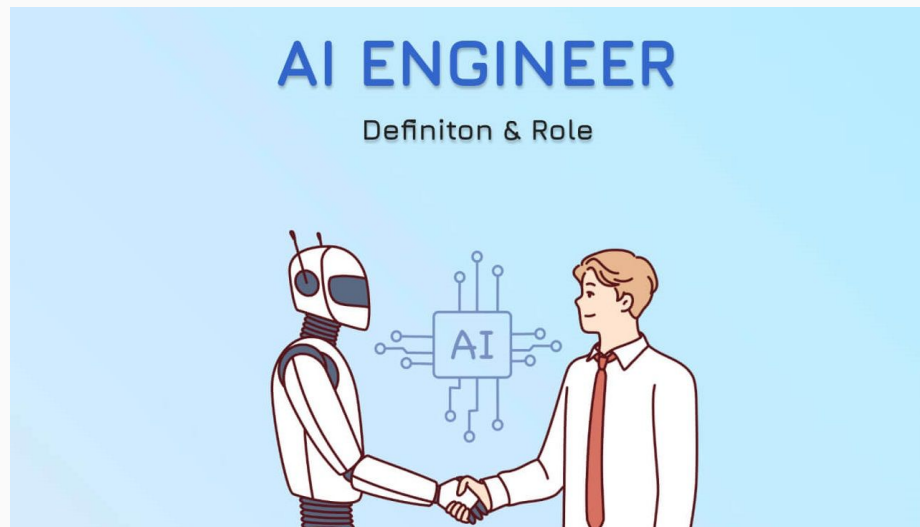
| Job                    | 25th percentile | 50th percentile | 75th percentile |
|------------------------|-----------------|-----------------|-----------------|
| AI/ML engineer         | \$134,000       | \$170,750       | \$179,500       |
| Cybersecurity engineer | \$118,500       | \$144,000       | \$190,750       |
| Data analyst           | \$96,250        | \$117,250       | \$138,500       |
| Data scientist         | \$121,750       | \$153,750       | \$182,500       |





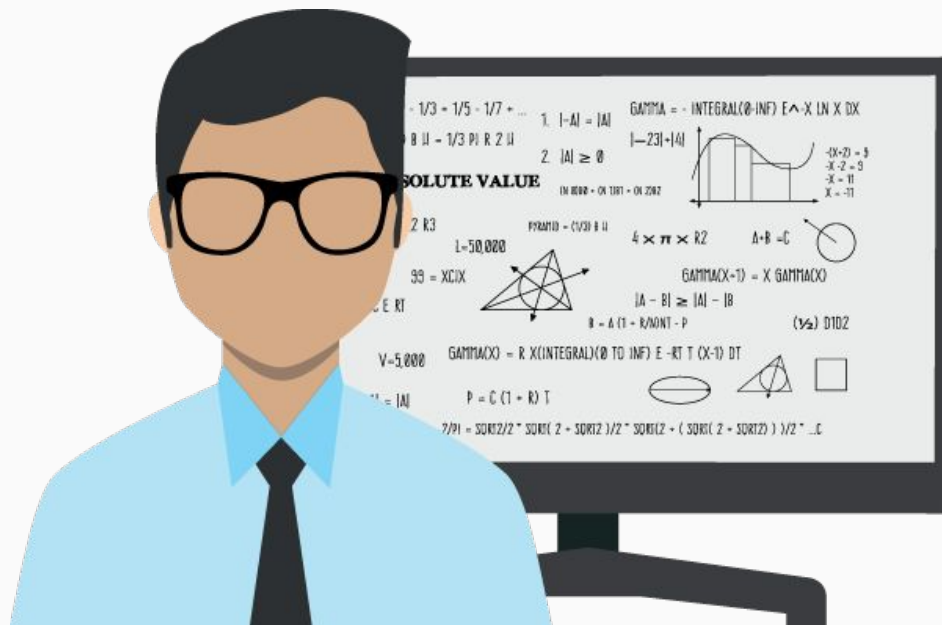
# Responsibilities: AI Engineer

- Develop AI-based applications
- Deploy AI models
- Integrate AI into systems
- Optimize performance
- Maintain AI pipelines



## Responsibilities: Data Scientist

- Data collection and cleaning
- Statistical analysis
- Model development
- Visualization and reporting
- Business insights generation



# Responsibilities: Machine Learning Engineer

- Build ML pipelines
- Train and optimize models
- Deploy models into production
- Monitor performance
- Scale solutions



**MACHINE  
LEARNING**

# Responsibilities: AI Researcher

- Conduct literature review
- Develop novel algorithms
- Publish research papers
- Experiment with new techniques
- Collaborate with academic and industry partners



# Responsibilities: AI Product Manager

- Define AI product strategy
- Coordinate between teams
- Understand user requirements
- Manage roadmap
- Ensure ethical AI use



# Research Conduction Process

1. Identify research problem
2. Literature review
3. Define research objectives
4. Methodology design
5. Data collection
6. Data preprocessing
7. Model development
8. Experimentation
9. Evaluation and validation
10. Results interpretation
11. Documentation
12. Publication or implementation

# Example AI Research Workflow

- Problem: Predict disease risk
- Data: Medical dataset
- Model: Machine learning classifier
- Evaluation: Accuracy, precision, recall
- Output: Decision support system
  -



# Why This Matters

- AI skills improve employability
- Automation saves time and cost
- AI supports research innovation
- Organizations need AI-ready professionals



# Key Takeaways

- AI is transforming every industry
- Learning AI is no longer optional
- Python enables powerful automation
- Multiple career paths available
- Structured research process is essential

# Thanks!

Contact us:

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